Ziad Ahmed

Senior Software Engineer | Backend & Full-Stack | AWS

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PROFESSIONAL SUMMARY

Senior Software Engineer who designs, builds and runs production systems end to end, from AWS infrastructure and CI/CD to Node.js/TypeScript backends and React/Next.js frontends. Backend specialist in NestJS, PostgreSQL, MongoDB and Redis with queue-based, event-driven architectures (BullMQ, AWS SQS/SNS). Shipped products across voice AI SaaS, HealthTech, e-commerce, F&B loyalty, real estate CRM and appointment booking, and currently lead a ~14-person engineering team across Egypt and India. Measurable wins include a 115× faster API, 98% less memory on large exports and ~$2,000/year cut from AWS spend. Teaching assistant in Software Architecture and Software Component Design at AASTMT.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Go, Python, Java, SQL

Backend: Node.js, NestJS, Express.js, RESTful APIs, GraphQL, gRPC, WebSockets, Webhooks

Frontend: React, Next.js, React Native

Cloud & DevOps: AWS (EC2, RDS, S3, SQS, SNS, SES, IAM, Amplify, AMI, ECS, ECR), Docker, Kubernetes, Terraform, GitHub Actions, CI/CD, NGINX, Cloudflare, Caddy

Databases: PostgreSQL, MySQL, MongoDB, Redis, SQLite; schema design, indexing, query optimization

Messaging & Queues: BullMQ, AWS SQS/SNS, RabbitMQ, Kafka

Architecture: Microservices, Event-Driven Architecture, Distributed Systems, Clean Architecture, DDD, SOLID, Design Patterns, Threat Modeling

Testing: Jest, Supertest, Mocha, Cypress; unit and integration testing

Integrations & Payments: Stripe, Twilio, Telnyx, Shopify, Firebase, APNs, Google APIs, BigQuery

AI Tooling: Claude Code, MCP, LLM pipelines; human-reviewed, AI-assisted development

PROFESSIONAL EXPERIENCE

Project Lead & Senior Software Engineer (Consultant) — Proven Solution Sep 2026 – Present

Contract · Remote | Real estate agency & CRM platform (ABHR)

- Lead the ABHR real estate CRM platform as Project Lead, coordinating a 14-person engineering team across Egypt and India and multiple time zones.
- Administer the project's AWS environment as one of two admins, giving the wider team scoped, non-admin access.
- Cut AWS spend by about $2,000/year: upgraded the RDS PostgreSQL version (~$150/month), deleted idle EC2 instances and released unused Elastic IPs.
- Resolved a production RDS outage caused by stale database connections leaking memory, and scaled RDS vertically to absorb load.
- Detected a leaked AWS access key, revoked it and replaced shared credentials with per-engineer IAM users scoped by least-privilege policies.
- Integrated Twilio, Telnyx, Facebook Lead Ads, Mailchimp, Microsoft Outlook, Google Calendar, Google Analytics and BigQuery.
- Built an automation rule engine on a database-backed job queue (DB poller and multithreaded workers), replacing a message broker and saving ~$20/month in infrastructure.
- Migrated a monorepo into 5 per-component repositories under a new GitHub organization, each with its own CI/CD pipeline, enabling independent builds and deployments.
- Cut deployment time by 70% (14 → 4.2 min) by parallelizing Java unit test execution and consolidating multiple compilation steps into a single build stage.

Senior Software Engineer (Consultant) — Nuxera Health Tech Aug 2026 – Present

Contract · Remote (KSA) | AI HealthTech · Next.js, NestJS, PostgreSQL, Redis, BullMQ

- Develop SERA, a real-time clinical transcription system, across a Next.js frontend and a NestJS backend on PostgreSQL, Redis and BullMQ.
- Migrated a tRPC codebase to standard REST APIs, removing unnecessary complexity and making the API usable by the desktop app, extension and other clients.
- Created a UI design system of shared, reusable components, replacing inconsistent AI-generated screens with one consistent product interface.

Senior Backend Engineer — Nugttah Jul 2025 – Jul 2026

Full-time | Loyalty & rewards platform for F&B · Node.js, Go, MongoDB, AWS

- Led an emergency AWS region migration (Bahrain to Mumbai) during regional escalation, with zero data loss.
- Made the Order List API 115× faster by optimizing MongoDB queries to restore index use, with no schema changes.
- Re-architected order exports to handle 100,000+ records without timeouts, using 98% less memory.
- Cut server startup time by 97% with async processing, non-blocking queues and lost-message handling.
- Resolved MongoDB replica-set race conditions between primary and secondary nodes, improving data consistency.
- Contained a Cloudflare security incident by moving users from the compromised service to internal infrastructure.

Senior Backend & Platform Engineer — Solid Ground Solutions LLC Jan 2025 – Apr 2026

Contract · Remote (US) | Voice AI SaaS · Next.js, Node.js, TypeScript, PostgreSQL, MongoDB, Redis, AWS, Stripe

- Sole backend engineer on an AI voice-agent outbound calling platform: built 153 REST APIs (~85K lines of TypeScript, 629 commits) covering campaigns, billing and real-time call monitoring.
- Designed a Redis-backed Deficit Round Robin dispatcher with atomic Lua-script leader election and per-user capacity limits, giving fair distributed call scheduling across multi-worker clusters.
- Built billing end to end: real-time credit deduction, Stripe Checkout webhooks and auto-recharge, automatic campaign pausing on low balance, OTP signup and automated Twilio subaccount provisioning.
- Refactored a 577-line campaign service into Clean Architecture, wrote the team engineering playbook and containerized the stack with Docker.
- Replaced Zapier with an internal workflow automation tool, saving about $300/month.
- Delivered multi-agent “Squads” call routing (AI-to-AI transfer chains) and a self-service voice-cloning library with S3-backed uploads.

Backend Engineer — Botit Sep 2023 – Sep 2025

Full-time | E-commerce & delivery platform · Node.js, MongoDB, Shopify

- Built real-time Shopify integrations and a notification service (APNs, Firebase) for event-driven data synchronization.
- Improved query performance by 30–42% and cut deployment time by 45% by optimizing GitHub Actions CI/CD.
- Wrote unit and integration tests, documented APIs in Swagger/OpenAPI, and applied threat modeling and input sanitization.
- Replaced Retool with a custom internal dashboard; took part in code reviews and led the engineering interview process.

PROJECTS

Naemn — Salon & Barber Booking Platform — Founder, Koala Studios 2026 – Present

naemn.app | React Native, Node.js, TypeScript, Express, MongoDB, Redis, AWS, Cloudflare

- Building an appointment-booking platform in Egypt for salons and home-visit barbers: customer mobile app (salon, date and time selection), partner app for salons and barbers, admin panel and dashboards.
- Shipped the customer mobile app with gamification (free haircuts, coupons) and safety verification for home visits; now building the iOS release and the partner app.
- Designed and run the AWS infrastructure: a single ARM EC2 instance behind Caddy with no SSH (Systems Manager only), S3 with CloudFront and signed links, SES email, and CI/CD from GitHub Actions to ECR.

KO-DB — Key-Value Database Engine in Go github.com/ZiadHaridy/KO-DB

- Built a key-value database engine from scratch in Go, implementing storage, retrieval and persistence internals.
- Implemented a compressed write-ahead log (WAL) for crash recovery and a custom TCP network layer for reads and writes.
- Sharded data across multiple clusters and published a YouTube walkthrough of the design and internals.

TEACHING EXPERIENCE

Teaching Assistant — AASTMT Feb 2026 – Present

Part-time | Software Engineering Department

- Teach practical sections in Software Component Design (design patterns) and Software Architecture (final-year course).
- Create bilingual (English/Arabic) materials, quizzes and grading rubrics; mentor students in Java, data structures, OOP, APIs and SQL.

EDUCATION

B.Sc. Computer Science Oct 2021 – Jun 2025

Arab Academy for Science, Technology & Maritime Transport (AASTMT)
GPA 3.72 / 4.0 · Ranked 3rd in cohort · Graduated with Excellent with Honors

LANGUAGES

Arabic (Native) · English B2